Customer Segmentation Prediction Using Machine Learning

P. Thrishal
Reg. No.: 241FA18450
Department of ACSE
Vignan's University
India

***Abstract*—Customer segmentation is an important machine learning application that helps organizations understand and categorize customers based on their demographic and behavioral characteristics. This project develops an end-to-end machine learning classification system for predicting customer segments. The dataset contains customer attributes such as age, gender, marital status, graduation status, profession, work experience, spending score, and family size. Exploratory data analysis and preprocessing techniques were applied to prepare the data for machine learning. Three classification algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were developed and evaluated using validation data. Among the evaluated models, Logistic Regression achieved the highest validation accuracy and was selected as the final model. The selected model was subsequently used to generate predictions for unseen customer data.**

***Index Terms*—Customer Segmentation, Machine Learning, Classification, Logistic Regression, Decision Tree, Random Forest, Exploratory Data Analysis**

I. Introduction

Customer segmentation is the process of dividing customers into different groups based on shared characteristics. It is widely used in marketing, customer relationship management, and business analytics to understand customer behavior and improve decision-making.

Machine learning can be used to automate customer segmentation by learning patterns from historical customer data. In this project, a supervised machine learning approach is used because the training dataset contains predefined customer segment labels.

The objective of this project is to develop an end-to-end machine learning system that can predict the customer segment of an unseen customer based on available demographic and behavioral attributes.

II. Problem Statement

The objective of this project is to predict the customer segment represented by the `Segmentation` variable using customer demographic and behavioral information.

The target variable contains four classes:

- Segment A
- Segment B
- Segment C
- Segment D

Since the target classes are already available in the training dataset, the problem is formulated as a supervised multi-class classification problem.

III. Dataset Description

The dataset used in this project was obtained from Kaggle. It contains customer demographic and behavioral attributes.

The main features used in the project include:

- **Gender** – gender of the customer.
- **Ever_Married** – marital status of the customer.
- **Age** – age of the customer.
- **Graduated** – whether the customer is a graduate.
- **Profession** – professional category of the customer.
- **Work_Experience** – years of work experience.
- **Spending_Score** – spending behavior category.
- **Family_Size** – number of family members.
- **Var_1** – additional categorical feature provided in the dataset.

The `ID` column was removed because it is an identifier and does not provide meaningful predictive information.

IV. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure and quality of the dataset before model development.

The following analyses were performed:

- Dataset dimensions and structure
- Missing value analysis
- Duplicate value analysis
- Target variable distribution
- Numerical feature distributions
- Outlier analysis using box plots
- Categorical feature distributions

A. Missing Value Analysis

Missing values were identified using the Pandas `isnull()` and `sum()` functions.

Numerical features were handled using median imputation, while categorical features were handled using mode imputation.

The median was selected for numerical features because it is less sensitive to extreme values. The most frequent category was used for categorical features.

B. Duplicate Analysis

Duplicate rows were checked in the training dataset. No duplicate rows were found.

C. Outlier Analysis

Box plots were used to identify potential outliers in numerical features such as Age, Work Experience, and Family Size.

Potentially unusual observations were investigated rather than automatically removed. Since values such as higher age or work experience can represent legitimate customers, no indiscriminate removal of outliers was performed.

V. Data Preprocessing

The following preprocessing operations were performed before model training.

A. Removal of Identifier

The `ID` column was removed because it uniquely identifies customers but does not represent a meaningful customer characteristic.

B. Handling Missing Values

For numerical features, missing values were replaced using the median value calculated from the training data.

For categorical features, missing values were replaced using the most frequent category.

C. Categorical Encoding

Categorical variables cannot be directly provided to most machine learning algorithms. Therefore, categorical features were converted into numerical representations using one-hot encoding.

This converts categories into binary indicator variables.

D. Feature Scaling

Standardization was applied to the features used by Logistic Regression.

The standardized value is calculated as:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where x is the original feature value, μ is the mean of the training feature, and σ is its standard deviation.

The scaler was fitted only on the training data and then used to transform validation and test data.

VI. Machine Learning Models

Three classification algorithms were implemented and compared.

A. Logistic Regression

Logistic Regression was used as a baseline classification model. It estimates the probability of a sample belonging to each customer segment and selects the class with the highest probability.

B. Decision Tree

Decision Tree classification uses a sequence of decision rules to divide the data into different groups. A maximum tree depth was specified to control model complexity and reduce the possibility of overfitting.

C. Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees. Multiple trees are trained and their predictions are combined to produce the final classification.

VII. Model Evaluation

The models were evaluated using the validation dataset.

The evaluation metrics used were:

- Accuracy
- Precision
- Recall
- F1-score

Accuracy is defined as:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \tag{2}$$

The F1-score combines precision and recall and is given by:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

A. Model Comparison

The validation accuracy obtained during the experiment was:

TABLE I
MODEL PERFORMANCE COMPARISON

Model	**Validation Accuracy**
Logistic Regression	51.67%
Decision Tree	49.32%
Random Forest	49.07%

Logistic Regression achieved the highest validation accuracy among the three evaluated models.

Therefore, Logistic Regression was selected as the final model.

VIII. Confusion Matrix

A confusion matrix was generated for the Logistic Regression model to analyze the classification performance across the four customer segments.

The rows of the confusion matrix represent the actual customer segments, while the columns represent the predicted segments.

Correct predictions occur along the main diagonal of the matrix, while off-diagonal values represent misclassifications.

IX. Final Prediction on Unseen Data

After selecting Logistic Regression as the final model, it was used to predict the customer segments of the unseen test dataset.

The test dataset contained 2,627 customer records.

The prediction results were saved to:

`outputs/test_predictions.csv`

The distribution of predicted segments was:

TABLE II
PREDICTED CUSTOMER SEGMENT DISTRIBUTION

Segment	Predicted Customers
A	752
B	374
C	707
D	794

X. Project Structure

The project was organized into separate directories to follow an end-to-end machine learning workflow.

```
MLOPS-prj1/
 configs/
 data/
    raw/
    processed/
 logs/
 models/
 notebooks/
 outputs/
 reports/
 src/
 tests/
 requirements.txt
```

The `notebooks` directory contains the exploratory and experimental work, while the `src` directory contains reusable Python code for preprocessing, training, and evaluation.

Trained model artifacts are stored in the `models` directory, and prediction results are stored in the `outputs` directory.

XI. Conclusion

This project developed an end-to-end machine learning classification system for customer segmentation. The dataset was explored and preprocessed by handling missing values, checking duplicates, analyzing outliers, encoding categorical features, and scaling numerical data where required.

Three machine learning models were evaluated: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression achieved the highest validation accuracy of approximately 51.67% and was therefore selected as the final model.

The selected model was applied to the unseen test dataset and generated predictions for all 2,627 customers. The predictions were saved as a CSV file for further use.

The results demonstrate the complete workflow from data understanding and preprocessing to model training, evaluation, model selection, and prediction on unseen data.

XII. Future Improvements

Future improvements could include:

- Testing additional classification algorithms.
- Applying systematic hyperparameter tuning when justified.
- Performing feature selection and feature engineering.
- Using cross-validation for more robust model evaluation.
- Investigating class imbalance and alternative evaluation metrics.
- Deploying the trained model as an API or web application.

References

[1] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[2] W. McKinney, "Data Structures for Statistical Computing in Python," in *Proceedings of the 9th Python in Science Conference*, 2010.

[3] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media.